V_{max} $\boxed{= 5V}$

$E_x = V_{max} = 5+5 = \boxed{10V}$

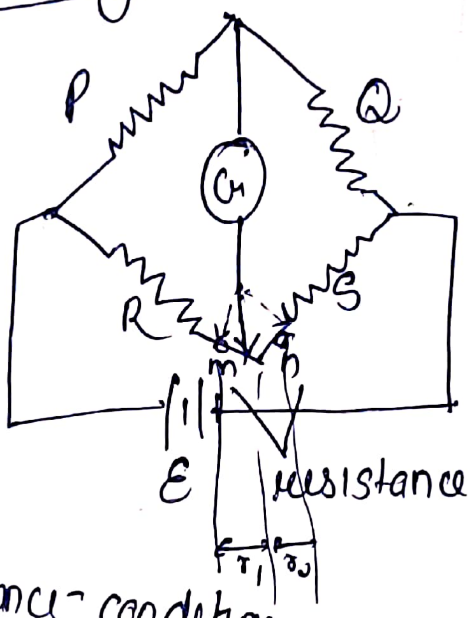
c) $10 = \frac{20 \times 100}{R+100}$

$R + 100 = 200$

$\boxed{R = 100\Omega}$

Measurement of Low Resistance ($\sim 1\Omega$)

Kelvin's Bridge



$\boxed{\frac{r_1}{r_2} = \frac{P}{Q}}$

we make arrangement

$\boxed{r_1 + r_2 = r}$

at balance condition

$\boxed{R + r_1 = \frac{P}{Q}(S + r_2)}$

$\frac{r_1}{r_1 + r_2} = \frac{P}{P+Q}$

$r_1 = \frac{Pr}{(P+Q)}$

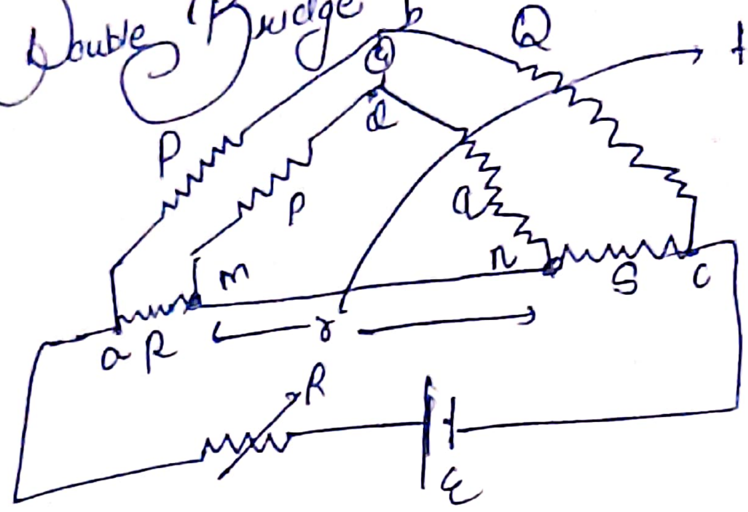
$r_2 = \left(\frac{Qr}{P+Q}\right)$

$R + \frac{Pr}{P+Q} = \frac{P}{Q}\left(S + \frac{Q}{P+Q}r\right)$

$\boxed{R = \frac{P}{Q}S}$

involvement of r is not affecting the overall equation

Kelvin Double Bridge



→ to be measured

$\begin{pmatrix} P, Q \\ p, q \end{pmatrix}$ → Two double arm

$$\frac{P}{Q} = \frac{p}{q} \text{ if } I_g = 0$$

$$R = \frac{P}{Q} \cdot S + \left(\frac{rQ}{P}\right)\left(\frac{P}{Q} - \frac{p}{q}\right)$$

$$= \frac{P}{Q} \cdot S + \frac{qr}{p+q+r}\left(\frac{P}{Q} - \frac{p}{q}\right)$$

$$E_{ab} = \frac{P\,E_{ac}}{P+Q}$$

$$E_{ac} = I\left[R + S + \frac{(p+q)r}{p+q+r}\right]$$

→ parallel combination b/w p

$$E_{amd} = I\left(R + \left(\frac{(p+q)r}{p+q+r}\right)\frac{p}{p+q}\right)$$

$$= I\left(R + \frac{pr}{p+q+r}\right)$$

In balance condition

$$E_{ab} = E_{amd}$$

$$\frac{P\,I}{P+Q}\left[R + S + \frac{(p+q)r}{p+q+r}\right] = I\left(R + \frac{pr}{p+q+r}\right)$$

$$\boxed{R = \frac{P}{Q}S}$$

addition of another branch is not affecting bridge equation

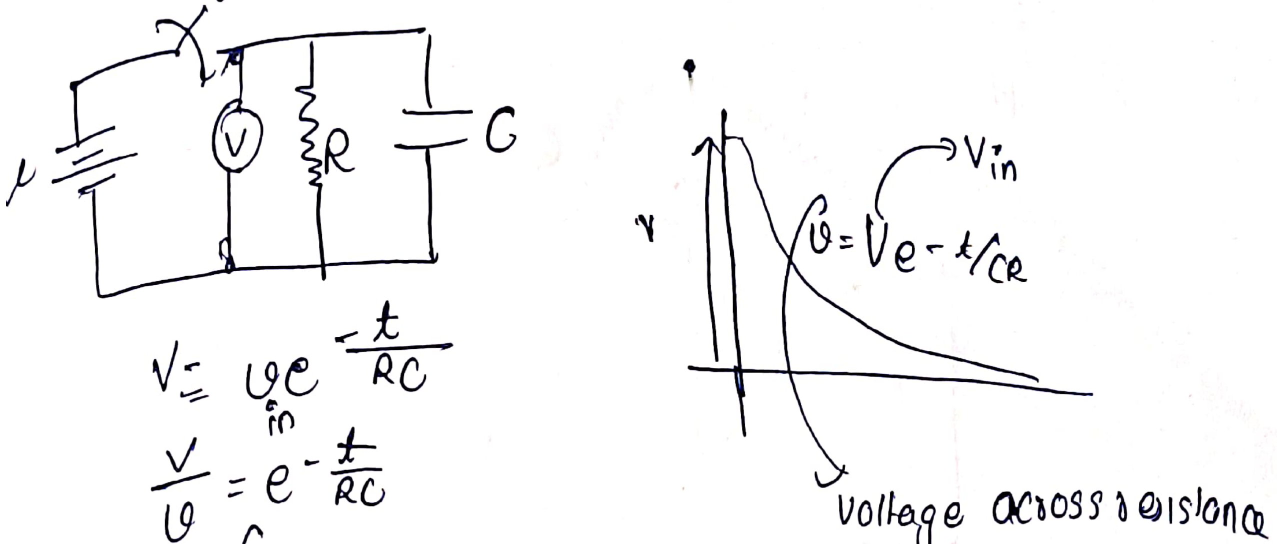
- It is desirable to keep the value of small r as low as possible in order to minimise the error in case

Here ... ratio $\frac{P}{Q} = \frac{p}{q}$

Measuring High Value of resistance ($\sim M\Omega$)

1. Loss of charge method
2. Mega Ohm Bridge

1) Loss of charge method



$$V = v_{in} e^{-\frac{t}{RC}}$$

$$\frac{V}{v} = e^{-\frac{t}{RC}}$$

$\boxed{\text{find R from here}}$ $R = \frac{t}{C \log_e\left(\frac{V}{v}\right)}$

$$R = \frac{0.4343\, t}{C \log_{10}\left(\frac{V}{v}\right)}$$

V → initial voltage

v → final voltage

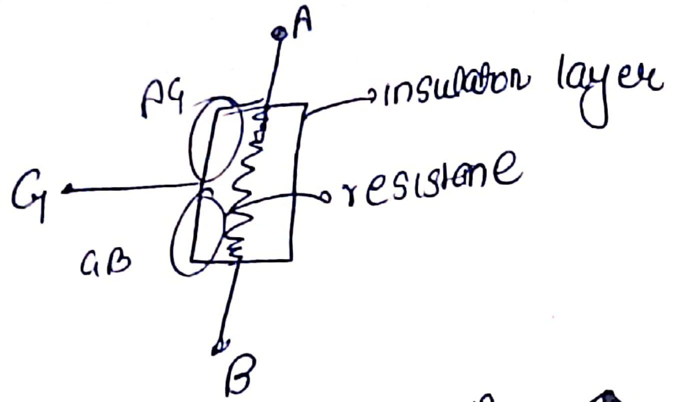
$$R = \frac{0.4343\, t}{C \log_{10}\left(\frac{V}{V-\varepsilon}\right)}$$

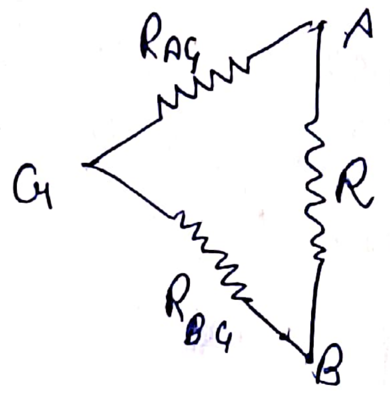
ε → residual voltage

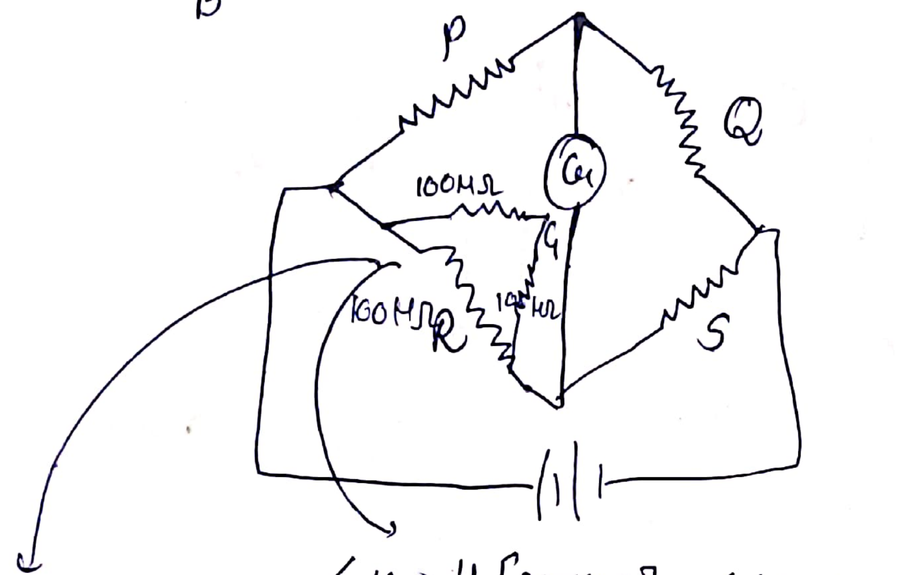
Limitations →

→ Requires resistance of very high leakage resistance similar to R

② Mega Ohm bridge →

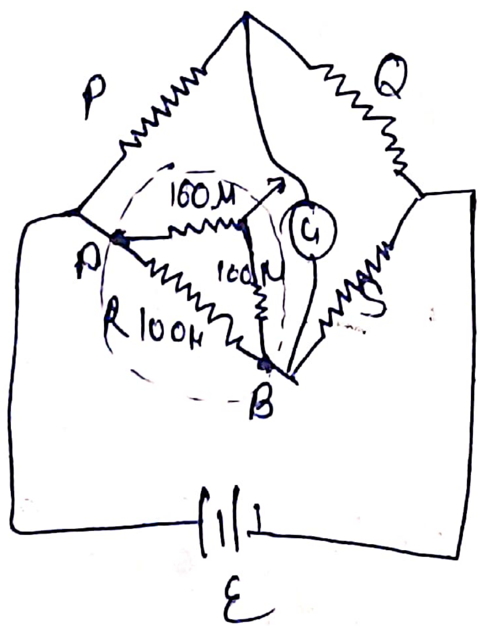






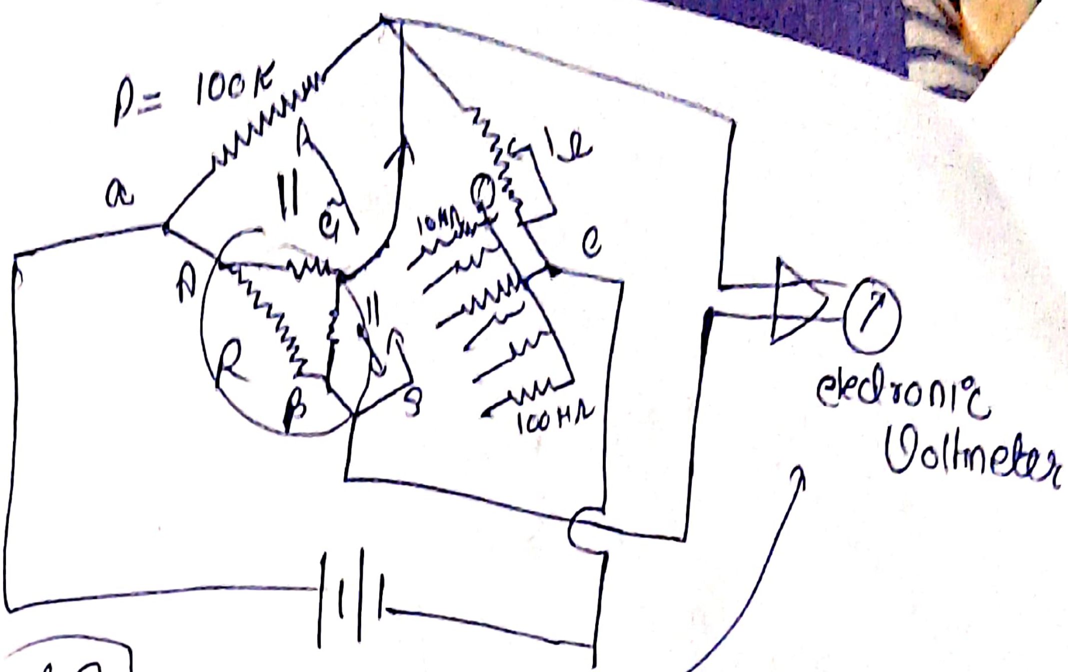
$< 100 \parallel [200 M\Omega] < 100 M\Omega$

measure $= \frac{100 \times 200}{300} = 67 M\Omega$ → So much error



→ In this case resistance BG is put in parallel with the galvanometer and thus it has no effect on balance because R_g is very low and will only affect the sensitivity of the galvanometer slightly the resistance R_{ag} is parallel with P and

this arrangement will give the correct value of R resistance



$\boxed{R = \frac{PS}{Q}}$ where it is 0

know

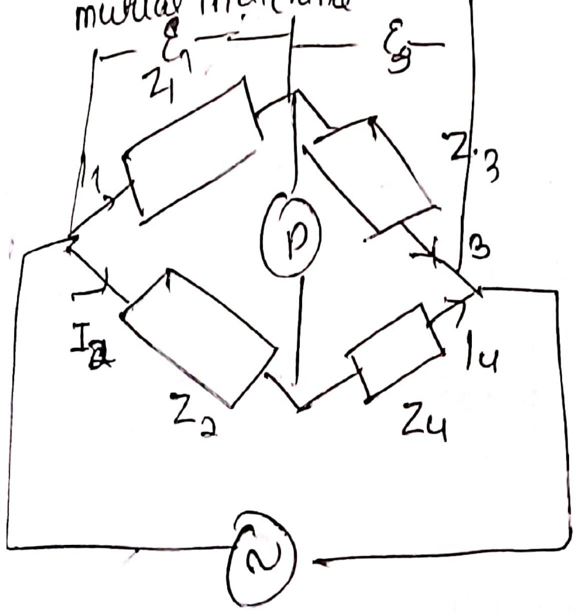
A.C. Bridges → capacitance

L, M, C, Q (measurement)

self inductance

Q - point

mutual indictance



during balance

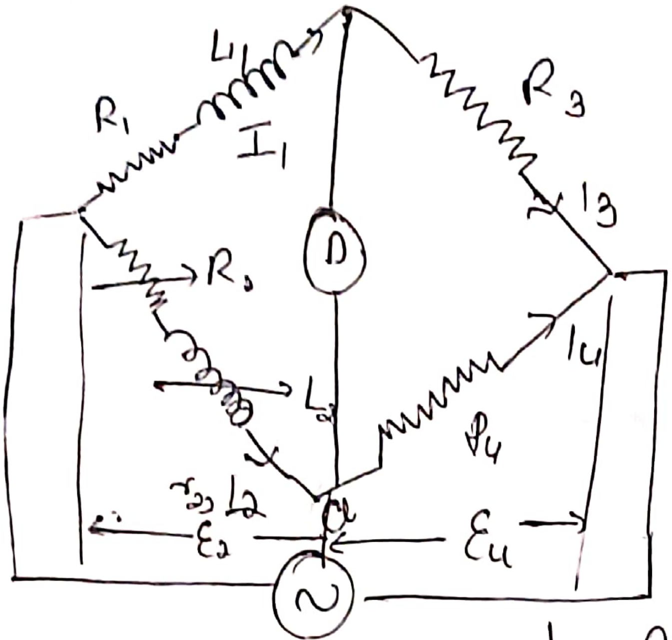
$$\frac{Z_1}{Z_2} = \frac{Z_3}{Z_4}$$

$$\underbrace{Z_1\angle\theta_1\, Z_4\angle\theta_4 = Z_2\angle\theta_2\, Z_3\angle\theta_3}$$

equate Real and imaginary terms and we will get answer

(a) Measurement of L

1) Maxwell inductance Bridge (unknown L, R)



L_1 = unknown inductance of Resistance R_1

L_2 → variable inductance of fixed resistance r_2

R_2 → variable resistance in series with L_2

R_3, R_4 → non inductive resistances

at balanced condition

$$Z_1 Z_4 = Z_2 Z_3$$

$$(R_1 + j\omega L_1) R_4 = (R_2 + r_2 + j\omega L_2) R_3$$

$$R_1 R_4 + j\omega L_1 R_4 = (R_2 + r_2) R_3 + j\omega L_2 R_3$$

real

$$\boxed{R_1 R_4 = (R_2 + r_2) R_3}$$

$$\boxed{\frac{R_1 R_4 - R_2 R_3}{R_3} = r_2}$$

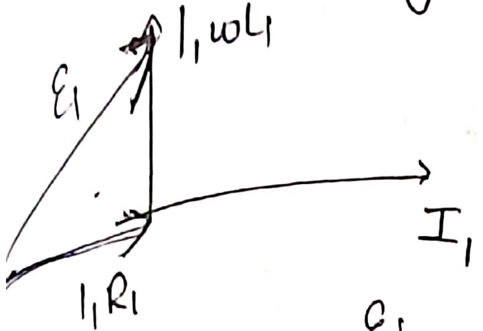
imaginary

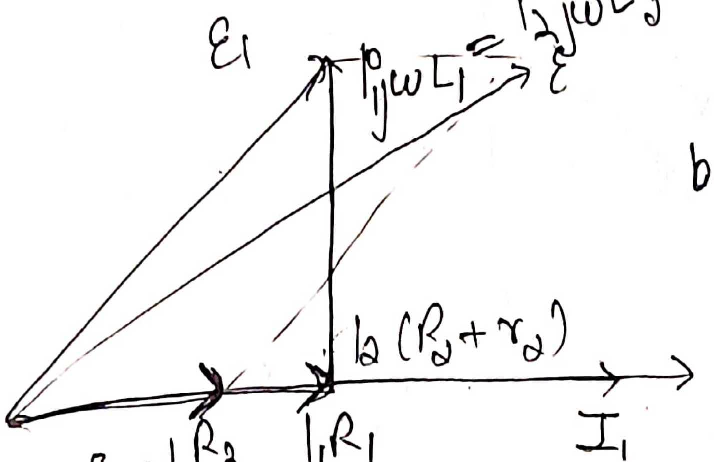
$$\cancel{\omega} L_1 R_4 = \cancel{\omega} L_2 R_3$$

$$\boxed{L_1 = \frac{R_3}{R_4} L_2}$$

$$\boxed{R_1 = \frac{R_3}{R_4}(R_2 + r_2)}$$

resistance of unknown branch





balanced condition

complex and real & seperate

$E_3 = I_1 R_3$

$E_4 = I_2 R_4$

$$\boxed{I_1 R_1 = I_2(R_2 + r)}$$

in balanced $\left.\begin{array}{l} E_1 = E_2 \\ E_3 = E_4 \end{array}\right\}$ in phasor

$E_1 = E_2$

$I_1 R_1 = I_2(R_2 + r)$

$I_1 \omega L_1 = I_2 \omega L_2$

$E_3 = E_4$

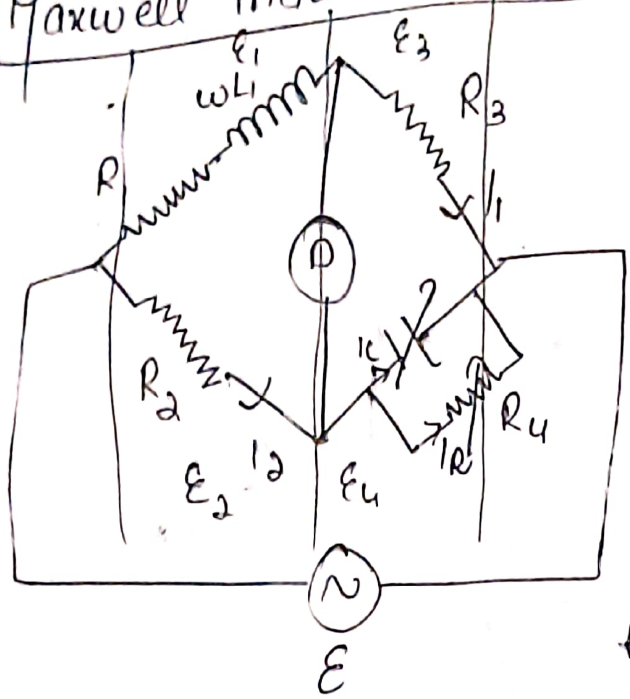
$I_1 R_3 = I_2 R_4$

$$\boxed{E_1 + E_3 = E}$$

for any bridge → $\mathcal{E}_1+\mathcal{E}_3=\mathcal{E}$

$\mathcal{E}_1=\mathcal{E}_2$ and $\mathcal{E}_3=\mathcal{E}_4$ } in any bridge

⊙ Maxwell inductance Capacitance Bridge



(R_1, L_1) → unknown

L_1 = unknown inductance

R_1 = " resistance associated with L_1

R_2, R_3, R_4 → non inductive resistance

C_4 ⇒ variable standard capacitance

$~~Z_1Z_4=Z_2Z_3~~$ $\quad Z_1Z_4=Z_2Z_3$

~~$(R_1+j\omega L_1)(R_4$~~

$$(R_1+j\omega L_1)\left(\frac{\frac{R_4}{j\omega C_4}}{R_4+\frac{1}{j\omega C_4}}\right)=R_2R_3$$

$$\boxed{R_1R_4+j\omega L_1R_4=R_2R_3+j\omega\frac{R_2R_3C_4}{R_4}}$$

$R_1R_4=R_2R_3$

$$\boxed{R_1=\frac{R_2R_3}{R_4}}$$

$$\boxed{L_1=R_2R_3C_4}$$

Q factor

$= \frac{\omega L_1}{R_1} = \frac{\text{reactance}}{\text{resistance}}$

$\boxed{Q = \omega C_4 R_4}$

advantages

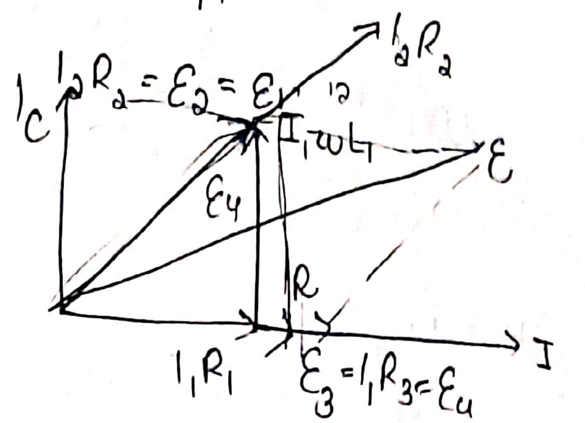
1] Two balance equations are independent

2] frequency term doesn't appear

⑥ phasor

$\mathcal{E}_1 = \mathcal{E}_2, \ \mathcal{E}_3 = \mathcal{E}_4$

$\mathcal{E}_1 + \mathcal{E}_3 = \mathcal{E} = \mathcal{E}_2 + \mathcal{E}_4$



$\boxed{I_2 = I_4}$

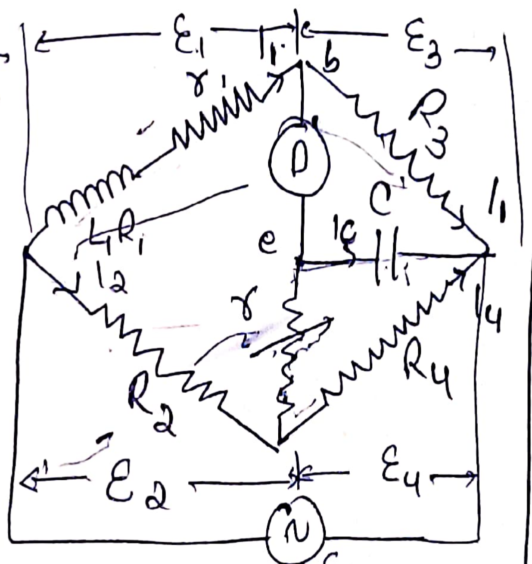
Anderson's Bridge

→ accurate measurement of L

→ advanced form of Maxwell's Inductance bridge

→ unknown L is compared with C_{fix}.

Construction



L_1 → self inductance to be measured

R_1 → resistance of self induct-

r_1 → resistance in series

r, R_2, R_3, R_4 → known non inductive resistances

C = standard fixed capacitor

• ~~Phasor~~ Diagram

At balance condition

$I_1 = I_3$ and $I_2 = I_c + I_4$

$I_1 R_3 = I_c \times \frac{1}{j\omega C}$ $(V_b = V_e)$

$\boxed{I_c = I_1 \omega C R_3}$

→ $\boxed{I_1(r_1 + R_1 + j\omega L_1) = I_2 R_2 + I_c r}$

$\boxed{I_c \left(r + \frac{1}{j\omega C}\right) = (I_2 - I_c) R_4}$

By subsituting I_c

$I_1(R_1 + r_1 + j\omega L_1) = I_2 R_2 + I_1 j\omega C R_3 r$

$I_1(r_1 + R_1 + j\omega L_1 - j\omega C R_3 r) = I_2 R_2$

and

$I_1(R_3 + j\omega R_3 R_4 + j\omega C R_3 r) = I_2 R_4 C$

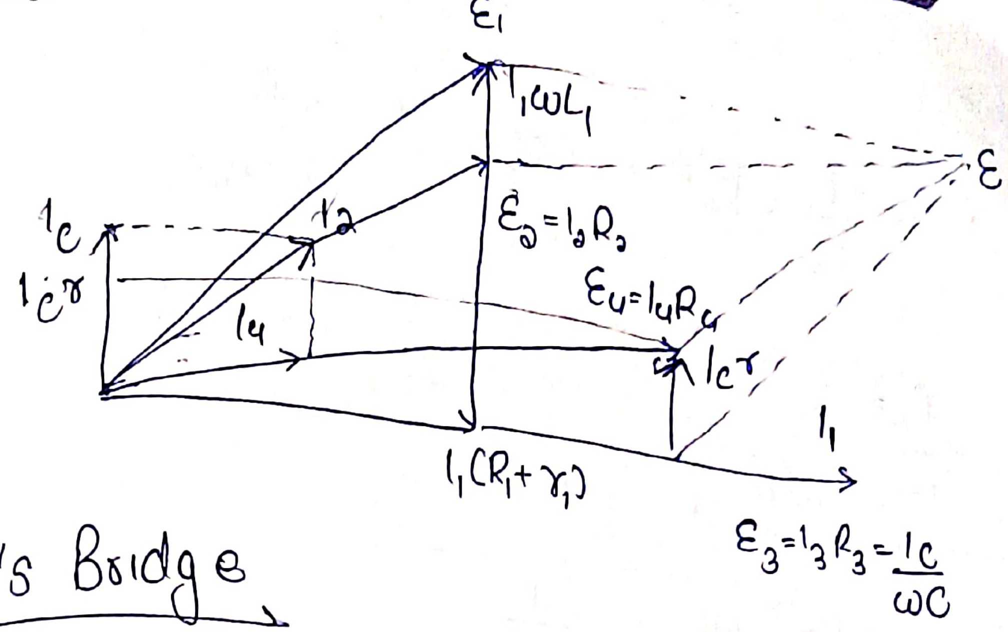
imaginary and real

→ $(r_1 + R_1) = \frac{R_1 R_2}{R_3}$

$\boxed{R_1 = \frac{R_2 R_3}{R_4} - r_1}$

$\boxed{L_1 = C \frac{R_3}{R_4}\left[r(R_4 + R_2) + R_2 R_4\right]}$

phasor →



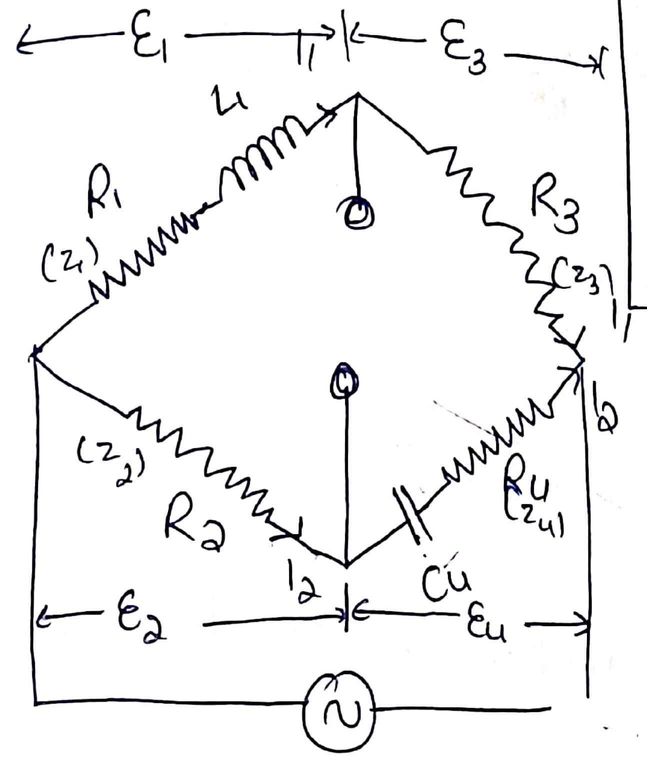
Hay's Bridge

→ determining L

→ advanced form of Maxwell's bridge

→ Maxwell → medium Q

→ Hay's → high Q



L_1 → unknown inductance with resistance R_1

R_2, R_3, R_4 → known non-inductive resistance

C_4 → standard capacitor.

$E_1 + E_3 = E$

$E_1 = E_2 \quad E_3 = E_4$

$$\frac{Z_1}{Z_2} = \frac{Z_3}{Z_4} \rightarrow \frac{R_1 + j\omega L_1}{R_2} = \frac{R_3}{R_4 - \frac{j}{\omega C_4}}$$

$$\left(R_1 R_4 + \frac{L_1}{C_4}\right) + j\left(\omega L_1 R_4 - \frac{R_1}{\omega C_4}\right) = R_2 R_3$$

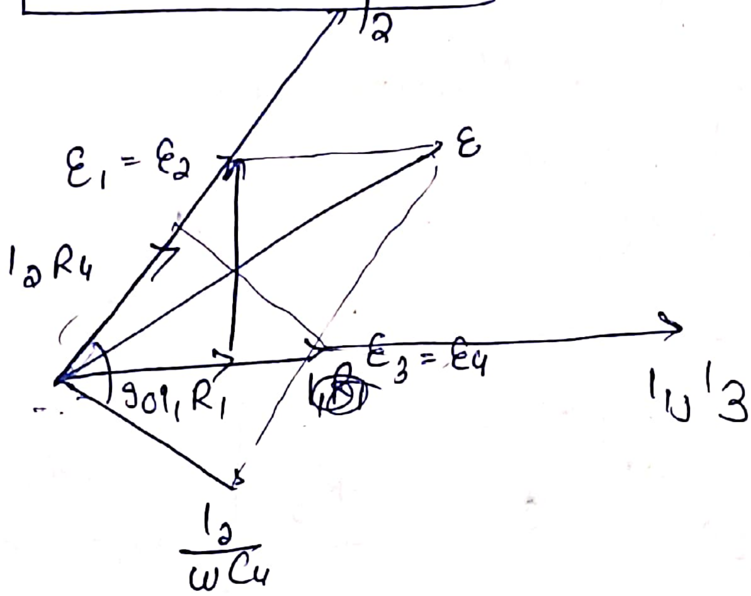
Seperating

$$R_1 R_4 + \frac{L_1}{C_4} = R_2 R_3 \quad \text{and} \quad L_1 = -\frac{R_1}{\omega^2 R_4 C_4}$$

$$\boxed{L_1 = \frac{R_2 R_3 C_4}{1+\omega^2 R_4^2 C_4^2}} \qquad \boxed{R_1 = \frac{\omega^2 C_4^2 R_2 R_3 R_4}{1+\omega^2 R_4^2 C_4^2}}$$

$$\boxed{Q = \frac{\omega L_1}{R_1} = \frac{1}{\omega^2 C_4 R_4}}$$

Phasor

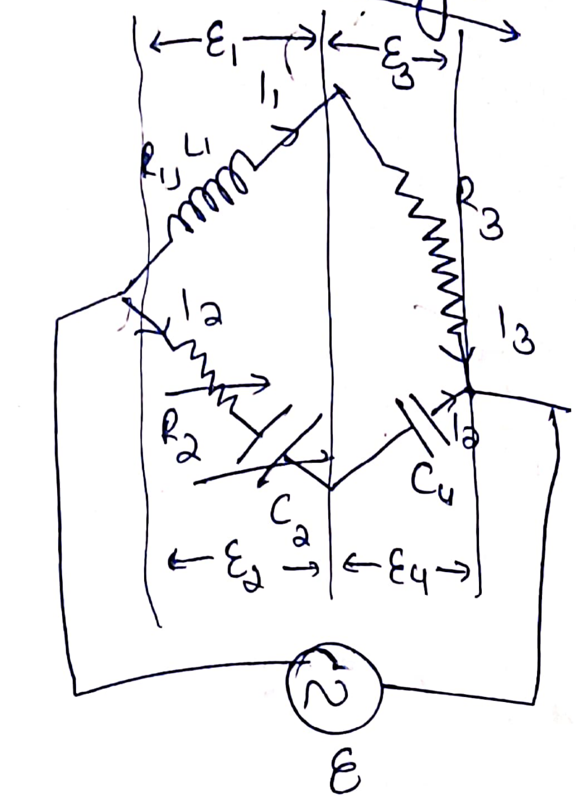


$$\boxed{Q \propto \frac{1}{\omega}}$$

→ frequency will affect Q

coil Q higher → this method will not be suitable

* Owen's Bridge



$$\frac{R_1 + j\omega L_1}{R_2 + \frac{j}{\omega C_2}} = \frac{R_3}{\frac{j}{\omega C_4}}$$

$$(R_1 + j\omega L_1)\left(\frac{-j}{\omega C_4}\right) = R_3\left(R_2 - \frac{j}{\omega C_2}\right)$$

$$\frac{-jR_1}{\omega C_4} + \frac{L_1}{C_4} = R_3 R_2 - \frac{jR_3}{\omega C_2}$$

$$\frac{L_1}{C_4} - j\frac{R_1}{\omega C_4} = R_3 R_2 - \frac{jR_3}{\omega C_2}$$

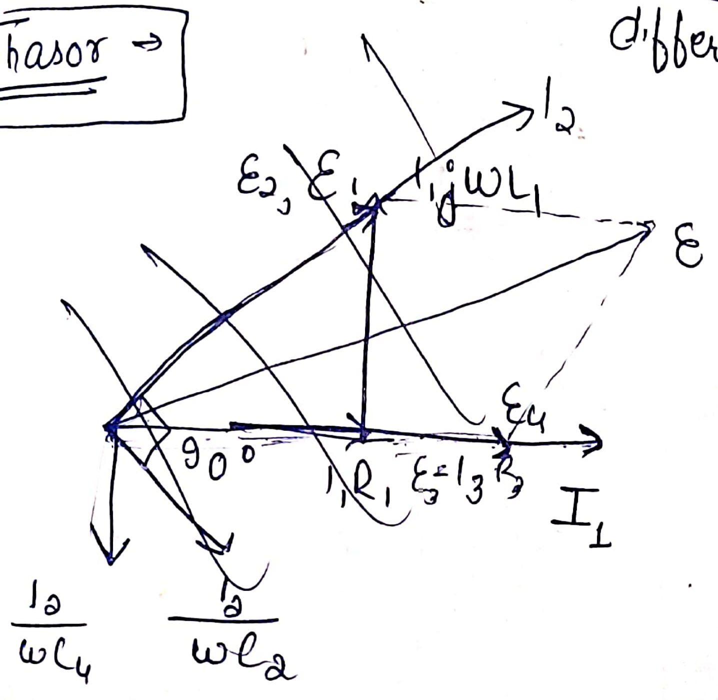
$$\frac{L_1}{C_4} = R_3 R_2$$

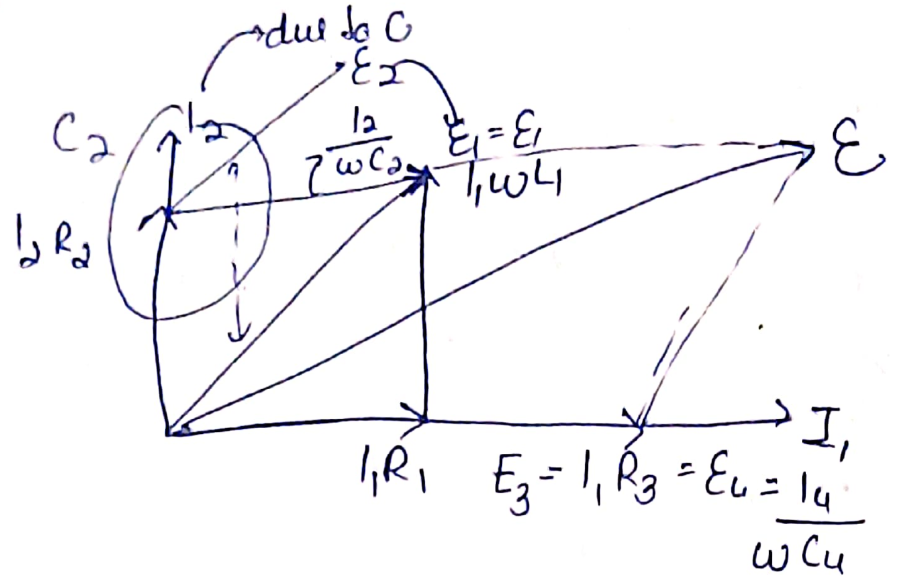
$$\boxed{L_1 = R_3 R_2 C_4}$$

$$\frac{R_1}{\cancel{\omega} C_4} = \frac{R_3}{\cancel{\omega} C_2} \qquad \boxed{R_1 = \frac{C_4 R_3}{C_2}}$$

different balanced cond^n easy

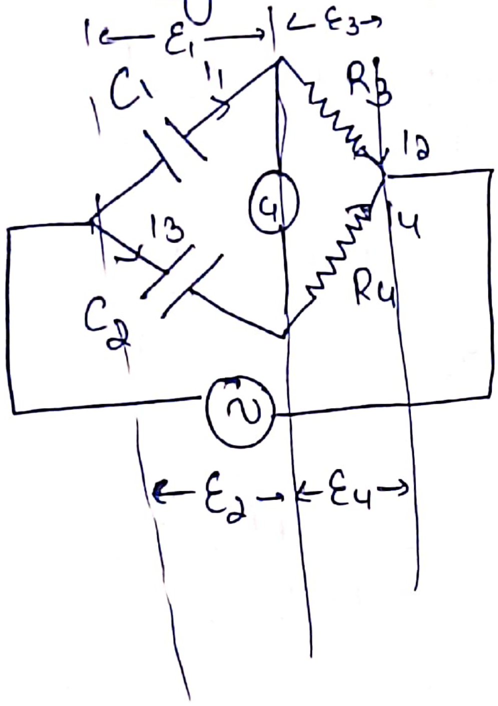
Phasor →





* Measurement of Capacitance

i) De sauty Bridge



C_1 → determined

C_2 → standard capacitance

R_3, R_4 → non inductive R

during balance

$$\frac{1}{j\omega C_1} R_4 = \frac{1}{j\omega C_2} R_3$$

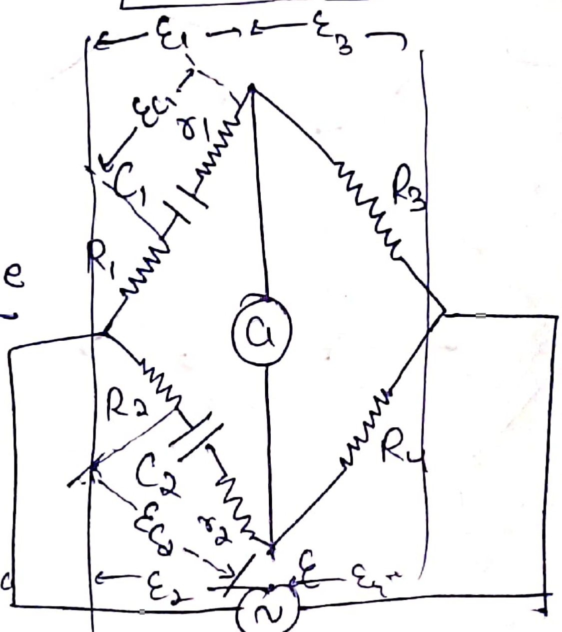
$$\boxed{C_1 = \frac{C_2 R_4}{R_3}}$$

C_1, C_2 → no dielectric loss if present

ii) modified de Sauty Bridge

r_1, r_2 cause of dielectric losses

R_1, R_2, R_3, R_4 } standard resistance



$$\left(R_1 + r_1 + \frac{1}{j\omega C_1}\right) R_4 = \left(R_2 + r_2 + \frac{1}{j\omega C_2}\right) R_2$$

$$\boxed{\frac{C_1}{C_2} = \frac{R_2 + r_2}{R_1 + r_1} = \frac{R_4}{R_3}}$$

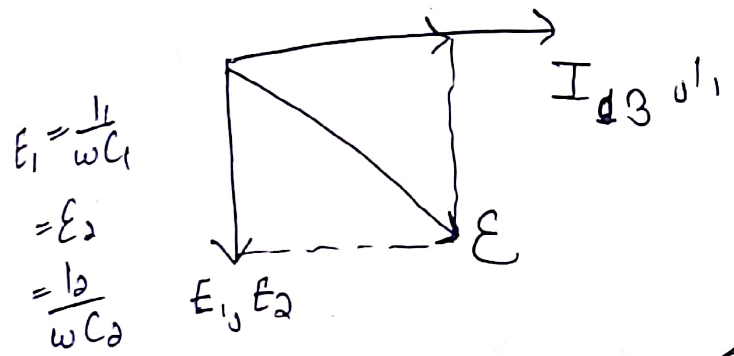
Dissipation factor

$$D_1 = \tan\delta_1 = \omega C_1 r_1$$
$$D_2 = \tan\delta_2 = \omega C_2 r_2$$

$$\frac{C_1}{C_2} = \frac{R_2 + r_2}{R_1 + r_1} = \frac{R_4}{R_3} \longrightarrow C_1 R_1 + C_1 r_1 = C_2 R_2 + C_2 r_2$$

$$D_1 - D_2 = \omega C_2 \left(\frac{R_1 R_4}{R_3} - R_2\right)$$

$E_3 = E_4 = I_3 R_3 = I_4 R_4$



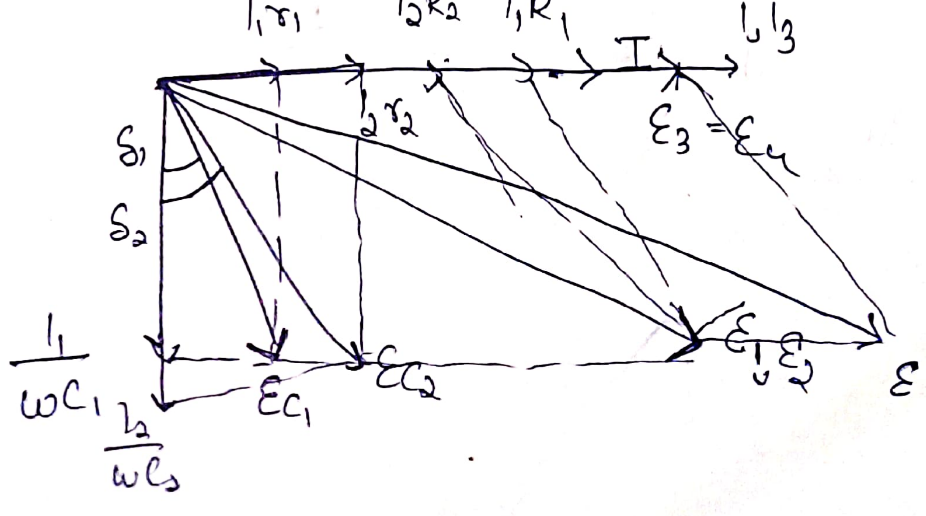
de-Sauty bridge

* modified

$$\frac{C_1}{C_2} = \frac{R_2 + r_2}{R_1 + r_1}$$

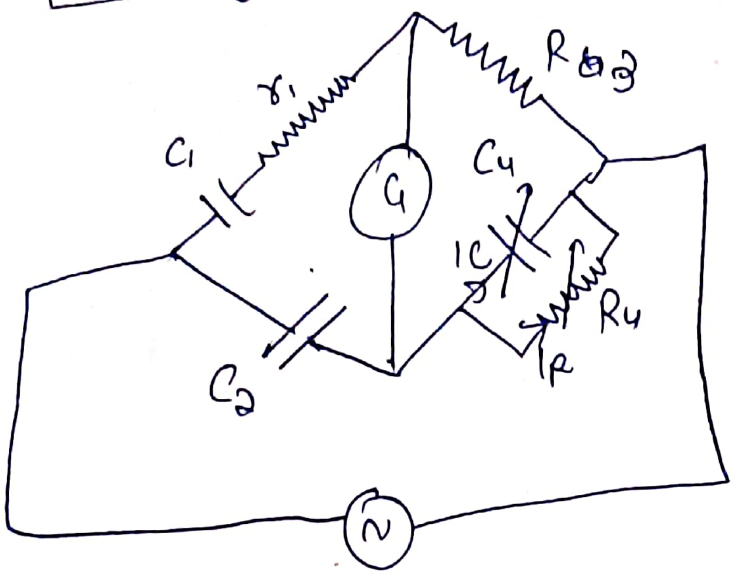
ratio

$(I_1 \to I_2)$ same phase



2) Sharing Bridge

Schering Bridge



$C_1 \rightarrow$ to be determined

Loss $r_1 \rightarrow$ series

$C_2 \rightarrow$ standard capacitor (Air)

$R_3, R_3 \rightarrow$ non inductive resistance

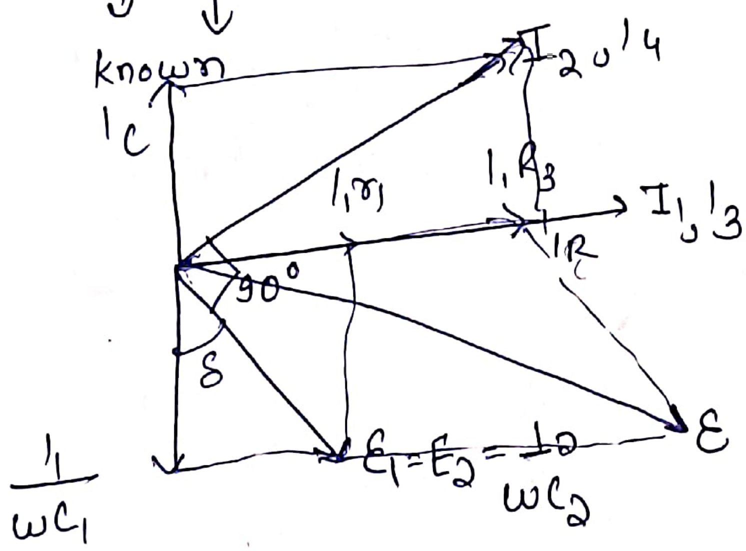
$C_4 \rightarrow$ Variable capacitor

$$\left(r_1 + \frac{1}{j\omega C_1}\right)\left(\frac{R_4}{1+j\omega C_4 R_4}\right) = R_3 \frac{1}{j\omega C_2}$$

$D_1 = \tan \delta_1 = \omega C_1 r_1$

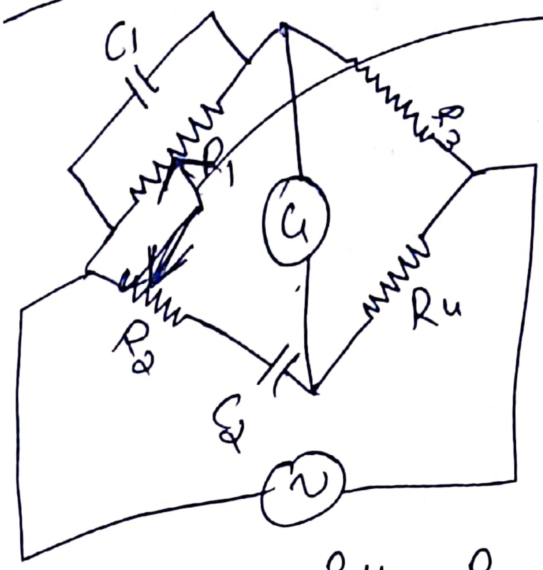
$$\boxed{r_1 = \frac{R_3 C_4}{C_2}}$$

$$\boxed{C_1 = C_2\left(\frac{R_4}{R_3}\right)}$$



frequency measurement

i) Wein's Bridge



→ mechanical linkage

→ same variation R_1 and R_2

$$\left(\frac{R_1}{1+j\omega C_1 R_1}\right) R_4 = R_3\left(R_2 - \frac{j}{\omega C_2}\right)$$

$$\frac{R_4}{R_3} = \frac{R_2}{R_1} = \frac{C_2}{C_1}$$

$$\boxed{\omega C_1 R_2 - \frac{1}{\omega C_2 R_1} = 0}$$

$$\omega = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

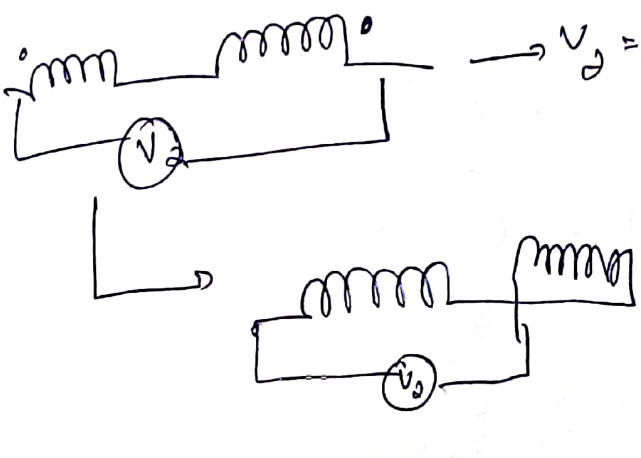
$$f = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$$

Measurement of mutual inductance

Case 1:

→ $V_1 = L_1 + L_2 + 2M$

Case 2:

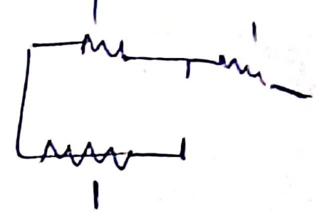


→ $V_2 = V_1 + L_2 - 2M$

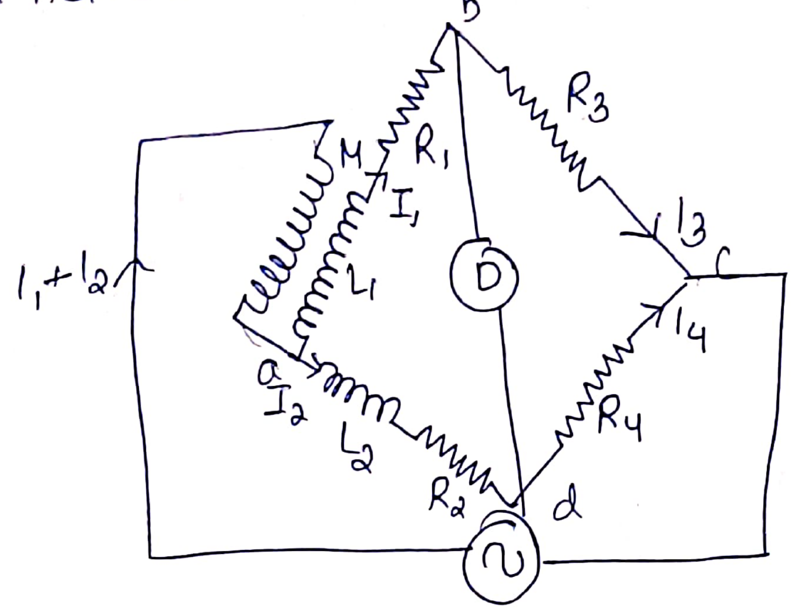
$$\boxed{M = \frac{1}{4}(V_1 - V_2)}$$

↓

simplest method



ii] Heaviside mutual inductance bridge



M = unknown M

L_1 = self inductance of secondary coil of mutual inductance

L_2 = known self inductance

R_1, R_2, R_3, R_4 = non inductive resistances.

$I_1 R_3 = I_2 R_4$

$V_{abc} = V_{adc}$

$$(I_1 + I_2) j\omega M + I_1(R_1 + R_3 + j\omega L_1) = I_2(R_2 + R_4 + j\omega L_2)$$

$$I_2\left(\frac{R_4}{R_3} + 1\right) j\omega M + \frac{R_4}{R_3} I_2 (R_1 + R_3 + j\omega L_1) = I_2(R_2 + R_4 + j\omega L_2)$$

$$\left(\frac{R_4}{R_3} + 1\right) j\omega M + \frac{R_4}{R_3}(R_1 + R_3 + j\omega L_1) = R_2 + R_4 + j\omega L_2$$

$$j\left(\frac{R_4}{R_3}\omega M + \omega L_1 \frac{R_4}{R_3}\right) = \omega L_2$$

$$\frac{R_4 M}{R_3} = L_2 - \frac{L_1 R_4}{R_3}$$

$$\boxed{M = \frac{R_3 L_2 - L_1 R_4}{R_3 + R_4}}$$

if $R_1 \ldots$

$$\boxed{M = \frac{L_2 - L_1}{2}}$$

This method can also be used for finding self inductance

$$\boxed{L_2 = \frac{M(R_3+R_4) + R_4 L_1}{R_3}}$$

Q] an Owen's bridge used to measure the properties of steel sheet (L, R) frequency is 2KHz at balance arm ab is the test arm bc $R_3 = 100\Omega$ arm cd = capacitor $C_4 = 0.1\mu F$ arm da $R_2 \to 834\Omega$ is series with capacitor having value of $0.124\mu F$ derive balance condition and calculate the effective impedence of specimen

$$Z \times X_C = R_3 \times Z_2$$

$$Z = \frac{100 \times \sqrt{834 + \left(\frac{10^6}{2\times\pi\times2\times10^3\times0.124}\right)}}{\frac{10^6}{0.1}}$$

$$Z = \frac{100 + \sqrt{834^2 + 79.57^2}}{\frac{10^6}{0.1}} = \frac{100 + 837.79}{\frac{10^6}{0.1}}$$

$$= \frac{937.79}{\frac{10^6}{0.1}}$$

$$\boxed{= 0.93\mu H}$$

Q.2] ab → choke coil R_1, L_1

bc → non inductive resistance R_3

cd → C_4 (micaconclensor) non inductive Resistance R_4 in se..

$\underbrace{\text{(micaconclensor)}}_{\text{dielectric}}$

da → non inductive resistence R_2

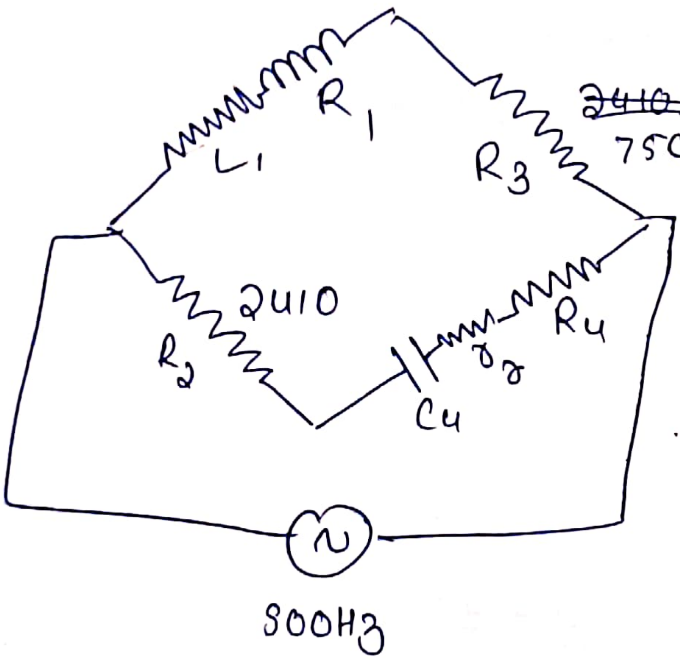
$f = 500\,Hz$ balance obtained $R_2 = 2410\,\Omega$

$R_3 = 750\,\Omega$

$C_4 = 0.35\,\mu F$

$R_4 = 64.5\,\Omega$

Series resistance of capacitor is $0.4\,\Omega$ calculate resistance and inductance of choke coil supply a and c.



$$(R_1 + j\omega L_1)\left((r_4 + R_4) - \frac{j}{\omega C_4}\right) = R_3 R_2$$

$$\left(R_1(r_4 + R_4) + \frac{L_1}{C_4}\right) + j\left(\omega L_1 (r_4 + R_4) - \frac{R_1}{\omega C_4}\right) = R_3 R_2$$

$$R_1(r_4 + R_4) + \frac{L_1}{C_4} = R_3 R_2$$

i) $\boxed{R_1\, 64.9 + L_1 (2857142) = 1807500}$

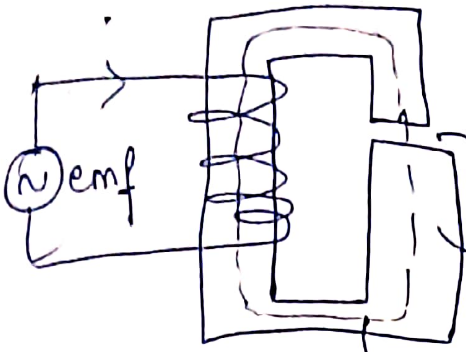
$$\omega(r_4 + R_4)L_1 - \frac{1}{\omega C_4} R_1 = 0$$

$\boxed{-909.45 R_1 + 203889 L_1 = 0}$ ii)

$R_1 = 141.1\,\Omega$

$L_1 = 0.63\,mH$

Magnetic Measurements



B → flux density / magnetic field strength

H → magnetising force

I → A

$\mathcal{E}$ → V

$\frac{\mathcal{E}}{I}$ → Resistance

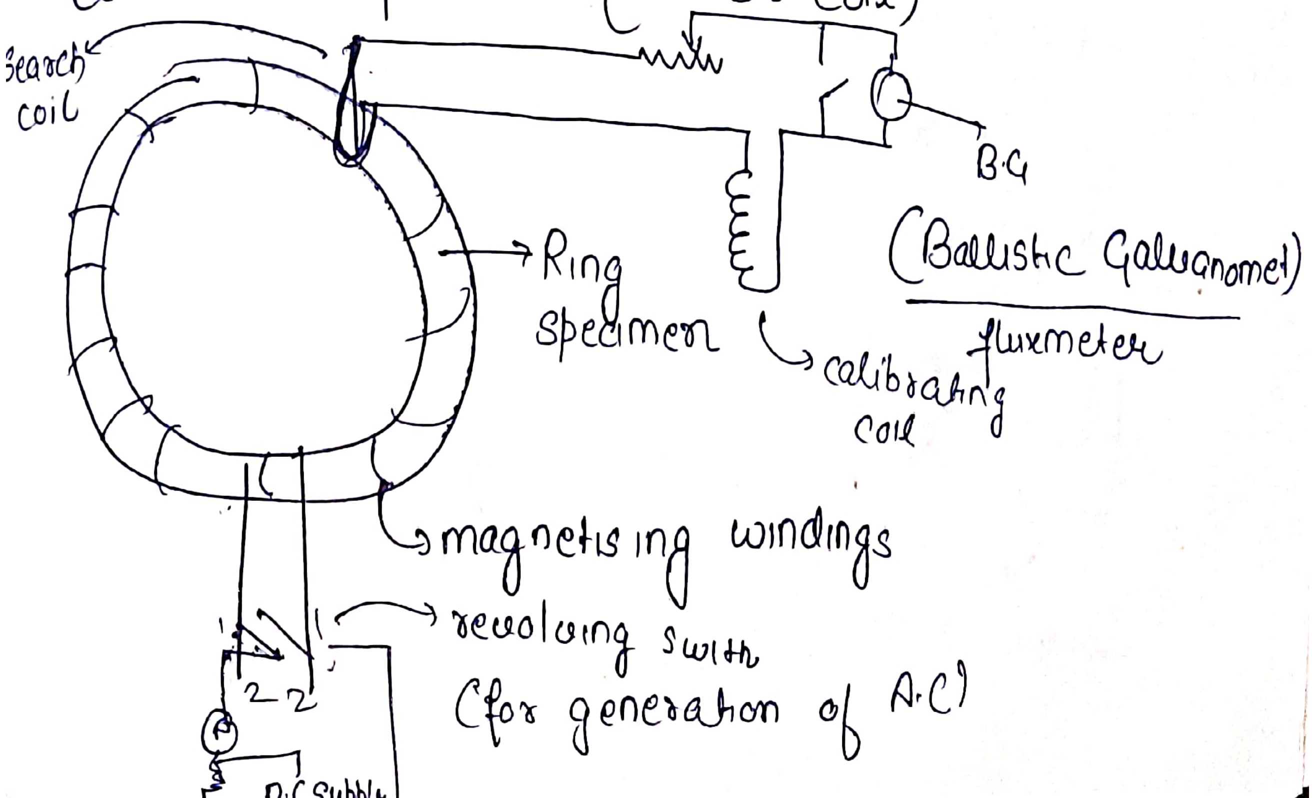
B → T / or Wb/m²

H → produce B → AT (mmf)

$\mathcal{R}$ → Relectuance → $\frac{H}{B}$

→ determine the B-H curve / hysterisis loop of a specimen

→ coil over specimen (search coil)



The current through the magnetising coil is reversed and the flux linkages in the search coil chang inducing an emf which sends a current through the fluxmeter or ballistic galvanometer indicating its value

let
$\phi \rightarrow$ flux linking the search coil
$R \rightarrow$ resistance of balistic galvanometer circuit
$N \rightarrow$ Number of turns in search coil
$t \rightarrow$ time taken to reverse the flux

$$e = N\frac{d\phi}{dt} \qquad e = \frac{2N\phi}{t}$$

flux changing 2 times in unit time

$$i = \frac{2N\phi}{tR} \qquad Q = \frac{2N\phi}{R}$$

Charge induced in the BG

$$= K_q \theta_1$$

$$\frac{2N\phi}{R} = K_q \theta_1$$

$$\boxed{\phi = \frac{K_q \theta_1 R}{2N}} \rightarrow B = \frac{\phi}{A}$$

linear scale

Cross sectional area search coil (magnetising)

Correction for the air flux

↳ loss of flux in air

$$B'A_s = BA_s + \mu_0 H (A_c - A_s)$$

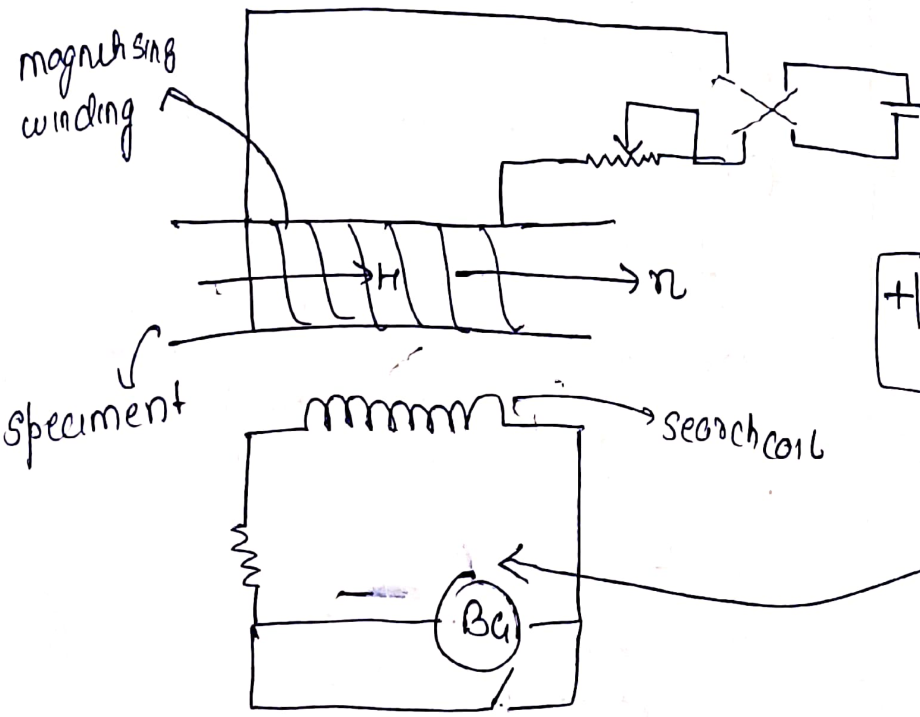
$B'A_s$ → observed flux

BA_s → true flux

$\mu_0 H (A_c - A_s)$ → lossed in air space between magnetising and search coil

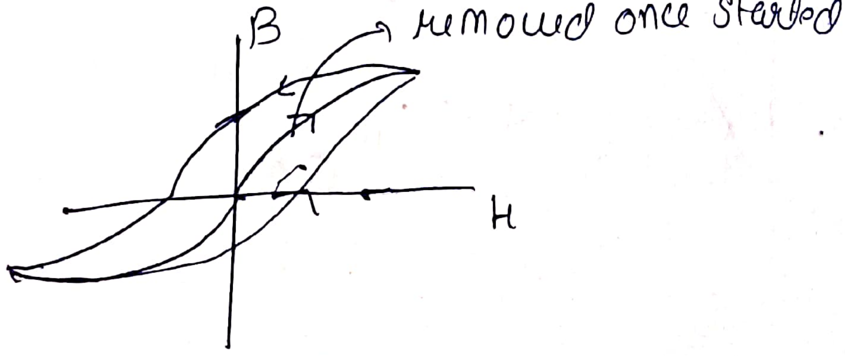
$$\boxed{B = B' - \mu_0 H \left(\frac{A_c}{A_s} - 1 \right)}$$

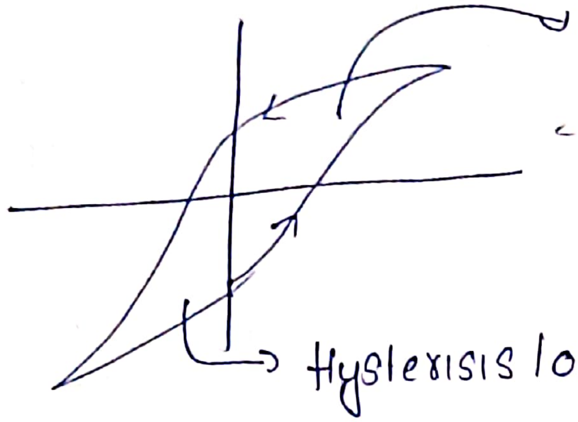
* Measurement of H (magnetising force)



$$\boxed{H = \frac{B_0}{\mu_0}}$$

BH curve

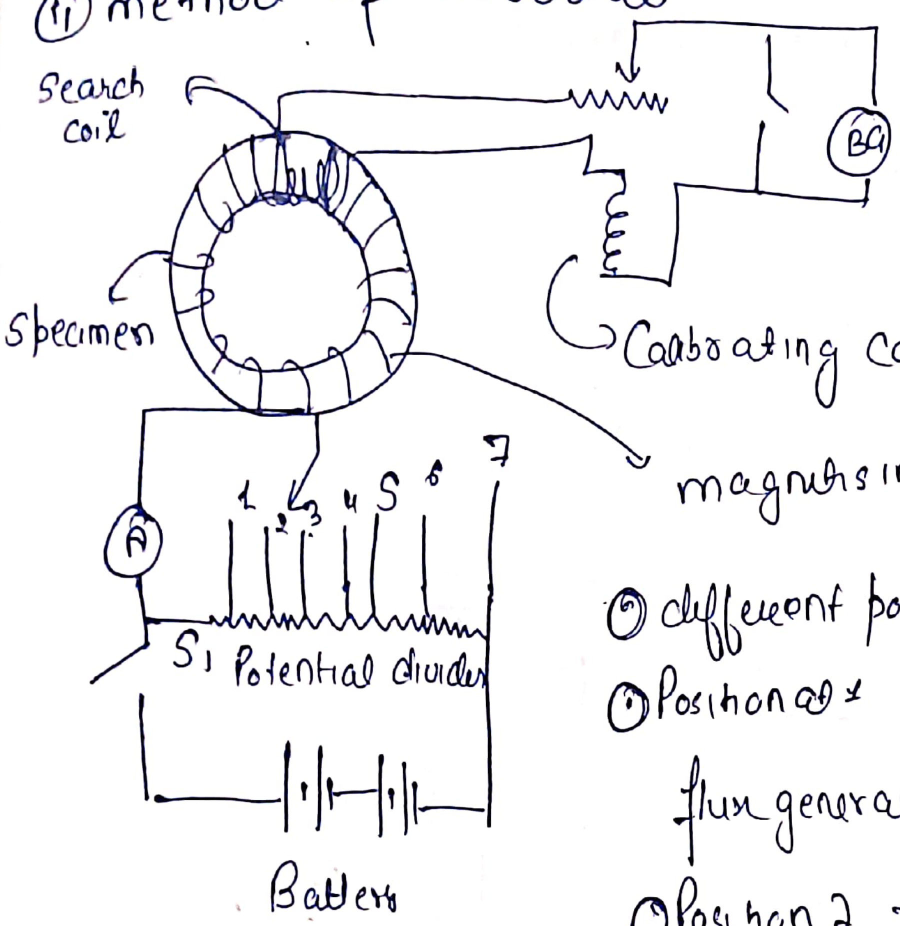




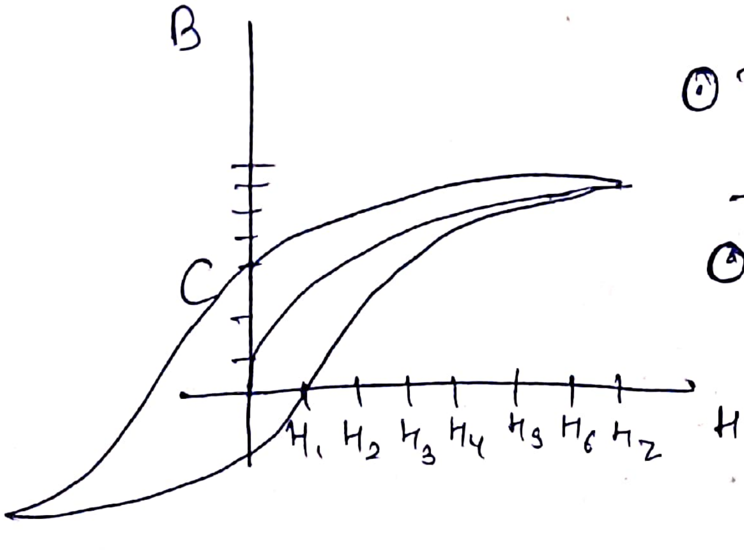
more the area of BH curve → more hysterises loss

entirely depends upon material

(i) Step by step method
(ii) method of reversal



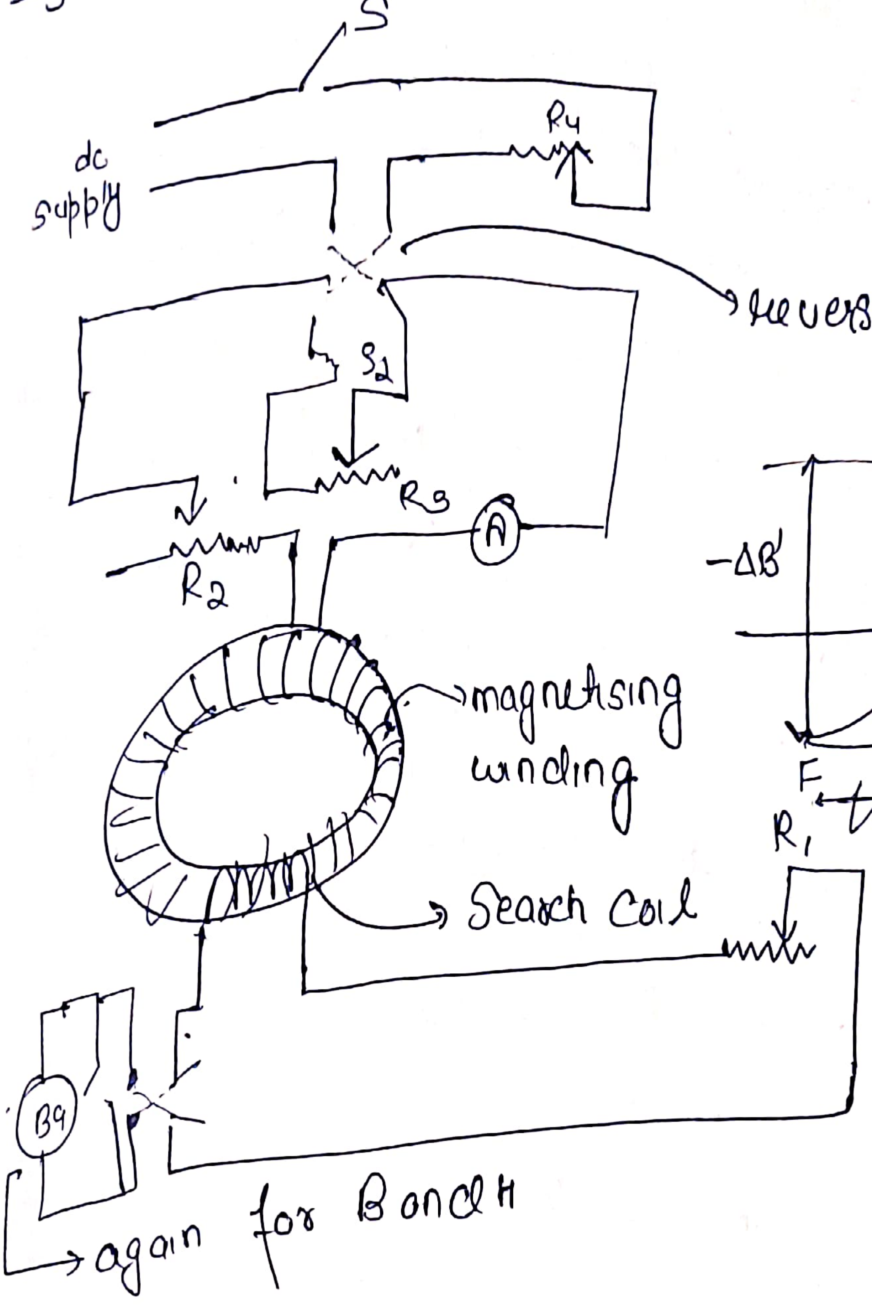
- different points
- Position at 1 → current I_1 → H_1
  flux generate → balistic galvanometer
- Position 2 → current I_2 → H_2 → ϕ_2
  → balistic galvanometer

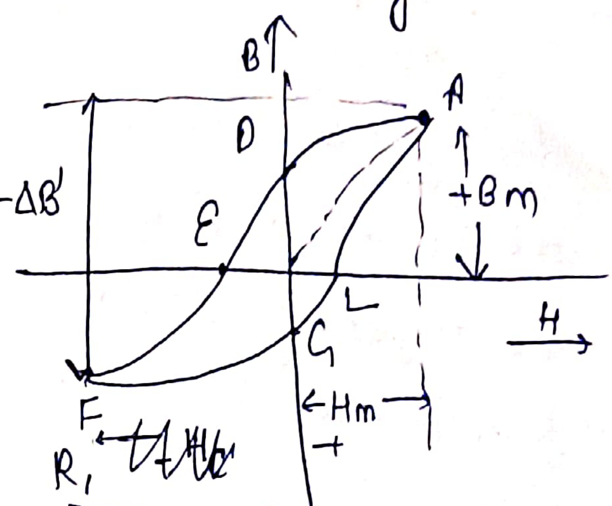


- Then again spring back to 7 from:
  →
- at point C reverse battery to reverse
- then again repeat the same process

①increase the potential value → increase neg potential →
dec → "
reverse ballose — sea ballon

2] Method of reversal →





A → C → D → E → F → G → L → A

① flux density is measured at each step from the maximum value $+B_m$ to some lower value. The connection R_1, R_2, R_4 in the magnetising winding and galvanometer winding R_3 a variable resistance which is connected across the magnetisation winding via switch S_2

→ R_2 and R_1 will be ... value of B and H →

Q] A ferromagnetic material is tested with help of a spermeater procuded with the coil wind over the specimen, this coil is known as B coil (search coil) and another coil is wound near the surface of the coil known as H coil thes two coil are connected to seperate balistic galvanometer having constant of 25×10^{-6} wb/turn/scale div the testing is done by reversing the current of the magnetising winding the reading of Balistic galv. connected to B coil → 100 div H coil → BG → 10 div (B, ϕ)

Coil	N	Area
B	100	$50 \times 10^{-6} m^2$
H	10,100	$5 \times 10^{-6} m^2$

i) find the flux density and magnetising force in the specimen

ii) find relative permeability of the specimen.

ϕ → flux N → number of turn

flux linkage $= N\phi$

due to reversal $= 2N\phi$

flux linkage with BG $= K_\phi \theta_1$

$$\boxed{2N\phi = K_\phi \theta_1}$$

$$\boxed{\phi = \frac{K_\phi \theta_1}{2N}}$$

→ R_2 and R_1 coil be
of B and H →

Q] A ferromagnetic material is tested with help of a spermeater procuded with the coil wind over the specimen, this coil is known as B coil (Search coil) and another coil is wound near the surface of the coil known as H coil these two coil are connected to seperate balistic galvanometer having constant of 25×10^{-6} wb/turn/scale div. the testing is done by reversing the current of the magnetising winding the reading of Balistic galv. connected to B coil → 100 div H coil → BG → 100 div
(B, ϕ)

Coil	N	Area
B	100	$50 \times 10^{-6} m^2$
H	10,100	$8 \times 10^{-6} m^2$

i) find the flux density and magnetising force in the specimen
ii) find relative permeability of the specimen.

ϕ → flux   N → number of turn

flux linkage $= N\phi$

due to reversal $= 2N\phi$

flux linkage with BG $= K_\phi \theta_1$

$$\boxed{2N\phi = K_\phi \theta_1}$$

$$\boxed{\phi = \frac{K_\phi \theta_1}{2N}}$$

$$B = \frac{K_\phi \theta_1}{2NA}$$

$$= \frac{\cancel{25} \times \cancel{10^{-6}} \times \cancel{100}}{2 \times \cancel{100} \times \cancel{50} \times \cancel{10^{-6}}} = 0.25\ Wb/m^2$$

4 coil → first B

$$B = \frac{K_\phi \theta_1}{2NA} = \frac{25 \times \cancel{10^{-6}} \times 10^{5}}{\cancel{2} \times 10{,}000 \times 5 \times \cancel{10^{-6}}}$$

$$B = 0.0025\ Wb/m^2$$

in 4 coil

$$\boxed{H = \frac{B}{\mu_0} = \frac{B}{4\times\pi\times10^{-7}} = 2000\ A/m \quad \boxed{2000\ AT}}$$

$$\mu_r = \frac{B}{\mu_0 H} = \frac{0.25}{4\times\pi\times10^{-7}\times2000} = \boxed{\mu_r = 100}$$

of the B coil